

DPP-8

- Q.1 Consider three vectors $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$. If $\vec{p}$, $\vec{q}$ and $\vec{r}$ denotes the position vector of three non-collinear points then the equation of the plane containing these points is
(A) $2x - 3y + 1 = 0$ (B) $x - 3y + 2z = 0$
(C) $3x - y + z - 3 = 0$ (D) $3x - y - 2 = 0$
- Q.2 The intercept made by the plane $\vec{r} \cdot \vec{n} = q$ on the x-axis is
(A) $\frac{q}{\hat{i} \cdot \vec{n}}$ (B) $\frac{\hat{i} \cdot \vec{n}}{q}$ (C) $(\hat{i} \cdot \vec{n})q$ (D) $\frac{q}{|\vec{n}|}$
- Q.3 If the distance between the planes
 $8x + 12y - 14z = 2$
and $4x + 6y - 7z = 2$
can be expressed in the form $\frac{1}{\sqrt{N}}$ where N is natural then the value of $\frac{N(N+1)}{2}$ is
(A) 4950 (B) 5050 (C) 5150 (D) 5151
- Q.4 A plane passes through the point P(4, 0, 0) and Q(0, 0, 4) and is parallel to the y-axis. The distance of the plane from the origin is
(A) 2 (B) 4 (C) $\sqrt{2}$ (D) $2\sqrt{2}$
- Q.5 If from the point P (f, g, h) perpendiculars PL, PM be drawn to yz and zx planes then the equation to the plane OLM is
(A) $\frac{x}{f} + \frac{y}{g} - \frac{z}{h} = 0$ (B) $\frac{x}{f} + \frac{y}{g} + \frac{z}{h} = 0$ (C) $\frac{x}{f} - \frac{y}{g} + \frac{z}{h} = 0$ (D) $-\frac{x}{f} + \frac{y}{g} + \frac{z}{h} = 0$
- Q.6 If the plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(k)$ with x-axis, then k is equal to
(A) $\sqrt{3}/2$ (B) $2/7$ (C) $\sqrt{2}/3$ (D) 1
- Q.7 The plane XOZ divides the join of (1, -1, 5) and (2, 3, 4) in the ratio $\lambda : 1$, then λ is
(A) -3 (B) -1/3 (C) 3 (D) 1/3
- Q.8 A variable plane forms a tetrahedron of constant volume $64 K^3$ with the coordinate planes and the origin, then locus of the centroid of the tetrahedron is
(A) $x^3 + y^3 + z^3 = 6K^3$ (B) $xyz = 6k^3$
(C) $x^2 + y^2 + z^2 = 4K^2$ (D) $x^{-2} + y^{-2} + z^{-2} = 4K^{-2}$
- Q.9 Let ABCD be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the area of triangles ABC, ACD and ADB be 3, 4 and 5 sq. units respectively. Then the area of the triangle BCD, is
(A) $5\sqrt{2}$ (B) 5 (C) $5/\sqrt{2}$ (D) 5/2

- Q.10 Equation of the line which passes through the point with p. v. (2, 1, 0) and perpendicular to the plane containing the vectors $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$ is
 (A) $\vec{r} = (2, 1, 0) + t(1, -1, 1)$ (B) $\vec{r} = (2, 1, 0) + t(-1, 1, 1)$
 (C) $\vec{r} = (2, 1, 0) + t(1, 1, -1)$ (D) $\vec{r} = (2, 1, 0) + t(1, 1, 1)$
 where t is a parameter
- Q.11 Which of the following planes are parallel but not identical?
 $P_1 : 4x - 2y + 6z = 3$
 $P_2 : 4x - 2y - 2z = 6$
 $P_3 : -6x + 3y - 9z = 5$
 $P_4 : 2x - y - z = 3$
 (A) P_2 & P_3 (B) P_2 & P_4 (C) P_1 & P_3 (D) P_1 & P_4
- Q.12 A parallelopiped is formed by planes drawn through the points (1, 2, 3) and (9, 8, 5) parallel to the coordinate planes then which of the following is not the length of an edge of this rectangular parallelopiped
 (A) 2 (B) 4 (C) 6 (D) 8
- Q.13 Vector equation of the plane $\vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ in the scalar dot product form is
 (A) $\vec{r} \cdot (5\hat{i} - 2\hat{j} + 3\hat{k}) = 7$ (B) $\vec{r} \cdot (5\hat{i} + 2\hat{j} - 3\hat{k}) = 7$
 (C) $\vec{r} \cdot (5\hat{i} - 2\hat{j} - 3\hat{k}) = 7$ (D) $\vec{r} \cdot (5\hat{i} + 2\hat{j} + 3\hat{k}) = 7$
- Q.14 The vector equations of the two lines L_1 and L_2 are given by
 $L_1 : \vec{r} = 2\hat{i} + 9\hat{j} + 13\hat{k} + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$; $L_2 : \vec{r} = -3\hat{i} + 7\hat{j} + p\hat{k} + \mu(-\hat{i} + 2\hat{j} - 3\hat{k})$
 then the lines L_1 and L_2 are
 (A) skew lines for all $p \in \mathbb{R}$
 (B) intersecting for all $p \in \mathbb{R}$ and the point of intersection is (-1, 3, 4)
 (C) intersecting lines for $p = -2$
 (D) intersecting for all real $p \in \mathbb{R}$
- Q.15 Consider the plane $(x, y, z) = (0, 1, 1) + \lambda(1, -1, 1) + \mu(2, -1, 0)$. The distance of this plane from the origin is
 (A) $1/3$ (B) $\sqrt{3}/2$ (C) $\sqrt{3/2}$ (D) $2/\sqrt{3}$